

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1709

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COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.(BL3)

Assessment Tool Weightage: 10%

QUESTIONS:2

Total Marks:20

Descriptive Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behaviour and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.
2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

Code of Conduct:

I P.Ramesh (192312443) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: P.Ramesh

RUBRICS FOR CALCULATION:

Criterion	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Understanding of Concepts	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions	5
Explanation	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation	5
Application	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis	4
Organization	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers	3
Accuracy	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps	2
Clarity	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand	1

1)ANS;

Intelligent Agents in a Smart Home Automation System;

A smart home automation system uses sensors, actuators, and artificial intelligence to control lighting, temperature, and security. The system receives real-time inputs such as motion detection, temperature, time of day, and user activities. It can also learn user preferences over time. Different types of intelligent agents can be used to implement this system.

Simple Reflex Agent

A simple reflex agent selects an action based only on the current percept using condition–action rules.

Examples:

- If motion is detected, turn ON the light.
- If temperature is above 28°C, turn ON the air conditioner.
- If a door is opened at night, activate an alarm.

Advantages:

- Simple and fast.
- Easy to implement.
- Suitable for immediate real-time responses.

Limitations:

- Does not remember past events.
- Cannot learn user preferences.
- Cannot handle partially observable situations effectively.

1. Model-Based Agent

A model-based agent maintains an internal model of the environment. It remembers previous states and uses them to make better decisions.

Example:

The system can remember:

- The user is currently away from home.
- The temperature has been increasing continuously.
- A door was opened while nobody was expected to be home.

Based on this internal state, it can make more intelligent decisions.

Advantages:

- Handles partially observable environments.
- Maintains information about the past.
- Can understand changes in the environment.

Limitations:

- More complex than a simple reflex agent.
- Requires memory and an accurate internal model.

2. Goal-Based Agent

A goal-based agent selects actions to achieve specific goals.

Possible goals include:

- Maintain comfortable temperature.
- Reduce energy consumption.
- Keep the house secure.
- Provide lighting according to user requirements.

Example:

If the goal is to save energy, the agent may:

- Turn OFF lights in empty rooms.
- Reduce air-conditioning when nobody is present.
- Use natural light during the daytime.

The agent evaluates possible actions and chooses those that help achieve its goals.

3. Utility-Based Agent

A utility-based agent selects the action that provides the highest overall utility or satisfaction.

In a smart home, several objectives may conflict:

- Maximum comfort may require more energy.
- Maximum energy saving may reduce comfort.
- Maximum security may cause unnecessary alarms.

Limitations:

- More complex to design.
- Requires suitable utility functions and accurate data.

Comparison of Agent Types

- Simple reflex component: Provides fast emergency responses, such as activating an alarm when suspicious motion is detected.
- Model-based component: Maintains information about occupancy, temperature history, and user behaviour.
- Goal-based component: Works toward goals such as comfort, security, and energy efficiency.
- Utility-based component: Balances competing objectives and selects the best overall action.
- Learning component: Learns user preferences over time, such as preferred temperature and lighting levels.

Conclusion;

A simple reflex agent is useful for quick reactions, but it cannot learn or plan. A model-based agent provides memory, while a goal-based agent works toward desired objectives. A utility-based agent is the most suitable for balancing multiple conflicting requirements. Therefore, a hybrid model-based, goal-based, and utility-based agent with learning capabilities is the best solution for a smart home automation system. It can provide comfort, reduce energy consumption, improve security, and adapt to individual user preferences.

2)ANS;

Robot Solving an Unknown Maze Using UCS and IDS;

A robot placed in an unknown maze must find a path from its starting position to the exit. Since the robot has no prior knowledge of the maze layout, it must explore the environment and search for a solution. Uninformed search algorithms such as Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can be used.

Uniform Cost Search (UCS)

Uniform Cost Search expands the node with the lowest path cost.

The robot always explores the least-cost path first.

Working

1. Start from the initial position.
2. Add possible movements to a priority queue.
3. Select the position with the lowest total path cost.
4. Continue expanding the lowest-cost position.
5. Stop when the exit is reached.

If all movements have equal cost, UCS behaves similarly to Breadth-First Search.

Advantages

- Complete: It will find a solution if one exists.
- Optimal: It finds the least-cost path when step costs are positive.
- Useful when different paths have different movement costs.

Disadvantages;

- Can consume large amounts of memory.
- May explore many unnecessary paths.
- Can be slow when the maze is large.

Complexity

- First search up to depth 0.
- If the goal is not found, search up to depth 1.
- Then search up to depth 2.
- Continue increasing the depth until the exit is found.

Advantages;

- Complete for finite branching factors.
- Finds the shallowest solution.
- Requires much less memory than UCS.
- Suitable for large search spaces.

Relationship with Intelligent Agents;

The design of the robot's intelligent agent affects how it makes decisions.

Simple Reflex Agent

A simple reflex agent reacts only to the current position and nearby obstacles.

Example:

- If the path ahead is blocked, turn left.
- If the exit is detected, stop.

Model-Based Agent;

A model-based agent maintains an internal map of the maze.

It can remember:

- Previously visited positions.
- Blocked paths.
- Available routes.
- Dead ends.

Utility-Based Agent;

A utility-based agent evaluates different paths based on factors such as:

- Shortest distance.
- Lowest energy consumption.
- Minimum risk.
- Maximum safety.

Conclusion;

the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. For an unknown maze, UCS is preferred when movement costs differ and finding the minimum-cost path is important. IDS is preferred when memory is limited and all movements have approximately equal cost.